

ANÁLISE E DESENVOLVIMENTO DE SISTEMAS

Computational Thinking

PROF. EDUARDO GONDO

Onde subir a API

- ▶ site render.com
- ▶ possibilidade de disponibilizar gratuitamente a API na internet
- ▶ o render pega diretamente o projeto do GitHub
- ▶ faça o cadastro no site e depois siga as imagens para criar a sua API

Escolhendo a API

The screenshot shows the FaaS6 console interface. At the top, there's a header with 'Jorge's workspace', 'Projects', a search bar, and buttons for '+ New', 'Upgrade', and a help icon. Below the header, the main heading is 'Create a new Service' with a 'Skip' link. A progress bar shows three steps: '1 Choose service', '2 Configure', and '3 Deploy'. A link 'Which to use' is on the right. The main area displays seven service options in a grid:

- Static Sites**: Static content served over a global CDN. Ideal for frontend, blogs, and content sites. [New Static Site →](#)
- Web Services**: Dynamic web app. Ideal for full-stack apps, API servers, and mobile backends. [New Web Service →](#) (highlighted with a blue arrow)
- Private Services**: Web app hosted on a private network, accessible only from your other Render services. [New Private Service →](#)
- Background Workers**: Long-lived services that process async tasks, usually from a job queue. [New Worker →](#)
- Cron Jobs**: Short-lived tasks that run on a periodic schedule. [New Cron Job →](#)
- Postgres**: Relational data storage. Supports point-in-time recovery, read replicas, and high availability. [New Postgres →](#)
- Key Value**: Managed Redis®-compatible storage. Ideal for use as a shared cache, message broker, or job queue. [New Key Value Instance →](#)

Configurando o GitHub

 Jorge's workspace 

New Web Service

Search  

[+ New](#) [Upgrade](#) 

Configure and deploy your new Web Service

 Choose service >  Configure >  Deploy [Need help? Docs](#) **Source Code**

Git Provider

Public Git Repository

Existing Image

Connect Git provider

Connect your Git provider to deploy from your existing repositories.

 GitHub

 GitLab

 Bitbucket

Name
A unique name for your web service.

example-service-name

Language
Choose the [runtime environment](#) for this service.

Node

Branch
The Git branch to build and deploy.

main

Configurando o GitHub

 Jorge's workspace 

New Web Service

Search 

+ New

Upgrade 

Configure and deploy your new Web Service

 Choose service >  Configure >  Deploy Need help? Docs **Source Code**

Git Provider

Public Git Repository

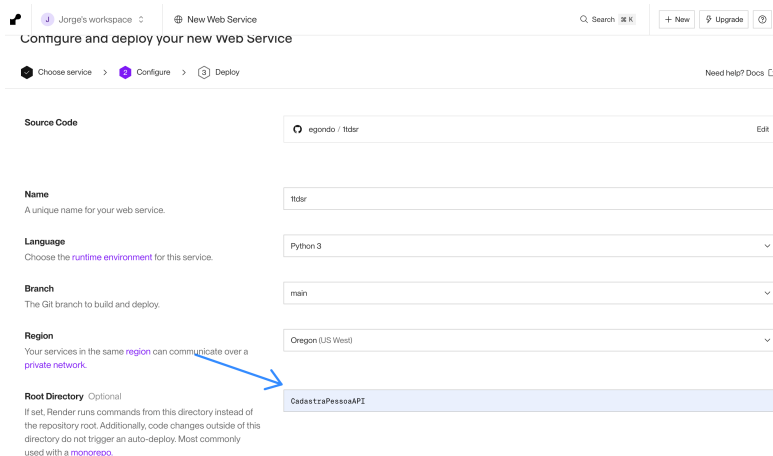
Existing Image

PR Previews and Auto-Deploy are available only for repositories configured with render.yaml

 Connect **Name**
A unique name for your web service.
Language
Choose the runtime environment for this service.
Branch
The Git branch to build and deploy.
Region
Your services in the same region can communicate over a

Configurando a API e o Flask

Escolhendo a pasta do projeto



Configure and deploy your new Web Service

Choose service > **Configure** > Deploy

Need help? Docs

Source Code egondo / ttdsr Edit

Name
A unique name for your web service. ttdsr

Language
Choose the runtime environment for this service. Python 3

Branch
The Git branch to build and deploy. main

Region
Your services in the same region can communicate over a private network. Oregon (US West)

Root Directory Optional
If set, Render runs commands from this directory instead of the repository root. Additionally, code changes outside of this directory do not trigger an auto-deploy. Most commonly used with a monorepo.

CadastrePessoaAPI

Configurando a API

- ▶ observe que as bibliotecas do nosso projeto precisam ser instaladas no render
- ▶ uma das formas é gerar o arquivo requirements.txt
- ▶ para isso, execute o comando `pip freeze > requirements.txt` no terminal
- ▶ abra o arquivo no visual studio code para checar se todas as bibliotecas estão presentes

Configurando a API

Jorge's workspace
New Web Service
Search
+ New
Upgrade

Root Directory Optional

If set, Render runs commands from this directory instead of the repository root. Additionally, code changes outside of this directory do not trigger an auto-deploy. Most commonly used with a [monorepo](#).

Build Command

Render runs this command to build your app before each deploy.

Start Command

Render runs this command to start your app with each deploy.

Instance Type

For hobby projects

Free	512 MB (RAM)	Upgrade to enable more features Free instances spin down after periods of inactivity. They do not support SSH access, scaling, one-off jobs, or persistent disks. Select any paid instance type to enable these features.
\$0 / month	0.1 CPU	

For professional use

For more power and to get the most out of Render, we recommend using one of our paid instance types. All paid instances support:

- Zero Downtime
- CDN Support

Starter	512 MB (RAM)	Standard	2 GB (RAM)
\$7 / month	0.5 CPU	\$25 / month	1 CPU
Pro	4 GB (RAM)	Pro Plus	8 GB (RAM)

```
CadastroPessoaAPI
```

```
CadastroPessoaAPI/ $ pip install -r requirements.txt
```

```
CadastroPessoaAPI/ $ flask run --host=0.0.0.0
```


Exercício

1. Suba um projeto no servidor render.com que acesse o banco de dados Oracle da FIAP

Referência Bibliográfica

- ▶ Puga e Rissetti - Lógica de Programação e Estrutura de Dados
- ▶ Ascêncio e Campos - Fundamentos da Programação de Computadores
- ▶ Forbelone e Eberspacher - Lógica de programação: a construção de algoritmos e estruturas de dados
- ▶ Documentação do Python - <https://docs.python.org/3.8/>
- ▶ Python Programming For Beginners: Learn The Basics Of Python Programming (Python Crash Course, Programming for Dummies) (English Edition). Kindle
- ▶ Python: 3 Manuscripts in 1 book: - Python Programming For Beginners - Python Programming For Intermediates - Python Programming for Advanced (English Edition). Kindle

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